

Long-acting Injectable (LAI) Antipsychotic Agents

Drug	Dosing	Pharmacokinetics	Additional Information
Haloperidol decanoate¹⁻⁵ (Haldol Decanoate®)	<u>Initiation:</u> 10-20x oral dose –OR– 10-15x po dose For doses >100mg: give up to 100mg, then administer remaining dose in 3-7 days <u>Maintenance:</u> 10-15x po dose q4 weeks (max dose: 450mg)	Half-life: 21 days Time to peak: 7 days Time to steady state: 3-4 doses • 3-4 months Duration: 2-4 weeks Metabolism & elimination: • Glucuronidation & CYP3A4 • 30% urine, 15% feces	<u>Storage:</u> Room temperature (20-25°C; 68-77°F) • Do not freeze or refrigerate Protect from light <u>Cost per dose:*</u> 100 mg/ml • \$131.53 <u>Administration:</u> Deep IM injection into gluteal muscle • 21G needle
Fluphenazine decanoate^{4,6-8} (Prolixin Decanoate®)	<u>Initiation & maintenance:</u> 6.25-25mg q2 weeks (max dose: 100mg) • Titrate in 12.5mg increments if >50mg is needed ***Note: 10mg oral ~ 12.5mg decanoate <u>Oral Overlap:</u> Minimum oral overlap w/ fluphenazine x1-2 weeks advised Recommend overlap until C _{ss} achieved	Half-life: ~14 days Time to peak: 8-10hrs Time to steady state: 4 doses • ~2 months Onset of action: 24-72 hrs • Peak effect 48-96hrs Duration: ~4 weeks Metabolism & elimination: • Glucuronidation • Modest CYP2D6	<u>Storage:</u> Room temperature (20-25°C; 68-77°F) • Do not freeze or refrigerate Protect from light <u>Cost per dose:*</u> 25 mg/ml • \$14.09 <u>Administration:</u> Gluteal muscle • ≥21G needle • Wet needle will make contents cloudy

*Cost data representative of acquisition cost from AmerisourceBergen as of 11/11/2014

Aripiprazole extended release⁹⁻¹¹ (Abilify Maintena®)	<u>Initial & maintenance:</u> 400mg IM once q4 weeks <u>Oral Overlap:</u> MUST overlap w/ any oral antipsychotic x14 days • Aripiprazole recommended to determine tolerance <u>Missed doses:</u> Second or third dose • 4-5 weeks late: Administer next dose ASAP • >5 weeks late: Administer oral aripiprazole x14 days w/ next scheduled dose Fourth or subsequent • 4-6 weeks late: Administer next dose ASAP • >6 weeks late: Administer oral aripiprazole x14 days w/ next scheduled dose <u>Dose Adjustments:</u> Adverse effects • Consider dose reduction to 300mg q4 weeks CYP3A4 inducers (e.g. carbamazepine): • Avoid use Strong CYP3A4 (e.g. ketoconazole) or CYP2D6 (e.g. fluoxetine) inhibitors >14 days • Reduce dose by 100mg - o If on 400mg: use 300mg q4 weeks - o If on 300mg: use 200mg q4 weeks Strong CYP3A4 and CYP2D6 inhibitors >14 days	Half-life: 300mg: 30 days 400mg: 40 days Time to peak: 5-7 days Time to steady state: 4-5 doses • ~4 months Onset of action: ~3 weeks Metabolism & elimination: • Hepatic CYP2D6 & CYP3A4 • 55% excreted in feces, 25% in urine	<u>Storage:</u> Room temperature (20-25°C; 68-77°F) • BRIEF excursions to 15°C (59°F) and 30°C (86°F) permitted Protect from light <u>Cost per dose:*</u> 300mg • \$1104.12 400mg • \$1472.17 <u>Administration:</u> Gluteal muscle • Rotate injection site • Allow ≥ 26 days between doses Reconstitute using provided SWFI & shake vigorously for 30 seconds Use provided needle • 21G x 1.5" • 21G x 2" in obese patients
--	--	--	---

*Cost data representative of acquisition cost from AmerisourceBergen as of 11/11/2014

	● If on 400mg dose: use 200mg q4 weeks ● If on 300mg dose: use 160mg q4 weeks Poor CYP2D6 metabolizers ● Reduce dose to 300mg q4 weeks		
Olanzapine pamoate¹²⁻¹⁵ (Zyprexa Relprevv®)	<u>Initial & maintenance:</u> Patients stable on 10mg po ● Initial: 210mg q2 weeks x4 doses –OR– 405mg q4 weeks x2 doses ● Maintenance: 150mg q2 weeks –OR– 300mg q4 weeks Patients stable on 15mg po ● Initial: 300mg q2 weeks x4 doses ● Maintenance: 210mg q2 weeks –OR– 405mg q4 weeks Patients stable on 20mg po ● Initial: 300mg q2 weeks ● Maintenance: 300mg q2 weeks <u>Oral Overlap:</u> None required <u>Dose Adjustments:</u> Geriatrics ● Consider initial dose of 150mg q4 weeks w/ caution Smoking ● Clearance may be increased 30-40%. May require dose adjustments after change in smoking status	Half-life: ~30 days Time to peak: ~7 days Time to steady state: 4-8 doses ● ~4 months Metabolism & elimination: ● Glucuronidation ● CYP1A2 & CYP2D6	REMS: Post-injection delirium/sedation syndrome ● Facility must be registered ● Monitor pts 3 hours after injection for sedation (possible coma) and delirium (agitation, anxiety, confusion, disorientation) <u>Storage:</u> Room temperature (20-25°C; 68-77°F) ● BRIEF excursions to 15°C (59°F) and 30°C (86°F) permitted Protect from light <u>Cost per dose:*</u> 210mg ● \$589.68 300mg ● \$842.40 405mg ● \$1137.24 <u>Administration:</u> Gluteal muscle ● MUST aspirate before injection to avoid vein Reconstitute using provided diluent, syringe, and needle ● 19G x 1.5" ● Obese: 19G x 2"

*Cost data representative of acquisition cost from AmerisourceBergen as of 11/11/2014

Risperidone 16-19 (Risperdal Consta®)	<u>Initial & maintenance:</u> 25-50mg IM q2 weeks (max dose: 50mg) - Make adjustments q4 weeks prn Note: 1mg oral ~ 12.5mg IM <u>Oral Overlap:</u> MUST overlap with any oral antipsychotic x3 weeks <u>Dose Adjustments:</u> None required - May consider starting at 12.5mg q2 weeks in renal impairment	Half-life: 3-6 days Time to peak: Drug released from microspheres at 3 weeks Time to steady state: 4 doses ~2 months Metabolism & elimination: - CYP2D6 - Urine 70%; feces 14%	<u>FDA max dose</u> 50 mg q2wks <u>Storage:</u> Refrigerate at 2-8°C (36-46°F) - Room temperature (20-25°C; 68-77°F) up to 7 days prior to administration Protect from light <u>Cost per dose:*</u> 12.5mg/2mL \$151.05 25mg/2mL \$302.05 37.5mg/2mL \$453.09 50mg/2mL \$604.13 <u>Administration:</u> Allow dose pack to come to room temperature before mixing Reconstitute using provided diluent, syringe, and needle(s) Alternate sites between injections - Deltoid: 21G x 1" needle - Gluteal: 20G x 2" needle

Paliperidone palmitate²⁰⁻²² (Invega Sustenna®)	<u>Initiation:</u> 234mg IM x1 dose, then 156mg IM 3-11 days later <u>Maintenance:</u> 39-234mg q3-4 weeks (oral dose dependent) If taking paliperidone ● 3mg po: 39-78mg ● 6mg po: 117mg ● 12mg po: 234mg If taking risperidone ● 1-2mg po: 39-78mg ● 4mg po: 117mg ● 6 mg po: 156 mg ● 9mg po: 234mg <u>Missed dose:</u> Second initiation dose (deltoid) ● <4 weeks late: Administer 2nd dose ASAP, then 3rd dose of 117mg 5 weeks later ● 4-7 weeks: Administer 156mg ASAP, then 156mg 1 week later, then resume as scheduled ● >7 weeks late: Reinitiate therapy Maintenance dose ● <6 weeks late: Administer dose ASAP ● 6-24 weeks late (deltoid) ○ <234mg: Administer normal dose ASAP, repeat in 1 week ○ 234mg: Administer 156mg ASAP, repeat in 1 week <u>Dose adjustments:</u> CrCl 50-79mL/min: 156mg x1, 117mg 1 week later; then 78mg q4 weeks	Half-life: 25-49 days Time to peak: 13 days Time to steady state: 4 doses ● ~4 months Onset of action: ~24 hours Metabolism & elimination: ● Limited CYP2D6 & CYP3A4	<u>Storage:</u> Room temperature (20-25°C; 68-77°F) ● Do not freeze or refrigerate ● BRIEF excursions to 15°C (59°F) and 30°C (86°F) permitted ● Protect from light <u>Cost per dose:*</u> 39mg ● \$293.49 78mg ● \$587.02 117mg ● \$880.52 156mg ● \$1174.07 234mg ● \$1761.08 <u>Administration:</u> 1st two loading doses = Deltoid muscle Maintenance = Deltoid or gluteal muscle Needle ● <u> </u> Deltoid (≥90kg): 22G x 1.5" ● <u> </u> Deltoid (≤90kg): 23G x 1" ● <u> </u> Gluteal: 22G x 1.5"

*Cost data representative of acquisition cost from AmerisourceBergen as of 11/11/2014

CrCl <50mL/min: Not recommended

Oral Overlap:
None required

References:

1. Haldol Decanoate (haloperidol) [package insert]. Titusville, NJ: Janssen Pharmaceuticals, Inc.; 2013.
2. Haloperidol. Lexi-Drugs Online. Hudson, OH: Lexi-Comp, Inc. Updated periodically. http://online.lexi.com/lco/action/doc/retrieve/docid/patch_f/7019/. Accessed November 5, 2014.
3. Haloperidol. DRUGDEX® System. Greenwood Village, CO: Thomson Healthcare. Updated periodically. http://www.micromedexsolutions.com/micromedex2/librarian/ND_T/evidencexpert/ND_PR/evidencexpert/CS/408C03/ND_AppProduct/evidencexpert/DUPLICATIONSHIELDSYNC/D0A387/ND_PG/evidencexpert/ND_B/evidencexpert/ND_P/evidencexpert/PFActionId/evidencexpert.DisplayDrugdexDocument?docId=269260&contentSetId=100&title=Haloperidol&servicesTitle=Haloperidol/. Accessed November 5, 2014.
4. Ereshefsky L, Toney G, Saklad SR, Anderson C, Seidel D. A loading-dose strategy for converting from oral to depot haloperidol. Hosp Community Psychiatry. 1993;44(12):1155-61.
5. Kreyenbuhl J, Buchanan RW, Dickerson FB, Dixon LB. The schizophrenia patient outcomes research team (PORT): updated treatment recommendations 2009. Schizophrenia Bulletin. 2010;36(1):94-103.
6. Prolixin Decanoate (fluphenazine) [package insert]. Weston, FL: Apotex Corp.; 2008.
7. Fluphenazine. Lexi-Drugs Online. Hudson, OH: Lexi-Comp, Inc. Updated periodically. http://online.lexi.com/lco/action/doc/retrieve/docid/patch_f/6934/. Accessed November 5, 2014.
8. Fluphenazine. DRUGDEX® System. Greenwood Village, CO: Thomson Healthcare. Updated periodically. http://www.micromedexsolutions.com/micromedex2/librarian/ND_T/evidencexpert/ND_PR/evidencexpert/CS/8EA71E/ND_AppProduct/evidencexpert/DUPLICATIONSHIELDSYNC/A14D6C/ND_PG/evidencexpert/ND_B/evidencexpert/ND_P/evidencexpert/PFActionId/evidencexpert.IntermediateToDocumentLink?docId=0354&contentSetId=31&title=FLUPHENAZINE&servicesTitle=FLUPHENAZINE/. Accessed November 5, 2014.
9. Abilify Maintena (aripiprazole) [package insert]. Tokyo, Japan: Otsuka Pharmaceutical Co. Ltd.; 2014
10. Aripiprazole. Lexi-Drugs Online. Hudson, OH: Lexi-Comp, Inc. Updated periodically. http://online.lexi.com/lco/action/doc/retrieve/docid/patch_f/6384/. Accessed November 5, 2014.
11. Aripiprazole. DRUGDEX® System. Greenwood Village, CO: Thomson Healthcare. Updated periodically. http://www.micromedexsolutions.com/micromedex2/librarian/ND_T/evidencexpert/ND_PR/evidencexpert/CS/98D3D5/ND_AppProduct/evidencexpert/DUPLICATIONSHIELDSYNC/0C234F/ND_PG/evidencexpert/ND_B/evidencexpert/ND_P/evidencexpert/PFActionId/evidencexpert.DisplayDrugdexDocument?docId=927028&contentSetId=100&title=Aripiprazole&servicesTitle=Aripiprazole/. Accessed November 5, 2014.
12. Zyprexa Relprevv (olanzapine) [package insert]. Indianapolis, IN: Lilly USA, LLC.; 2011.
13. Olanzapine. Lexi-Drugs Online. Hudson, OH: Lexi-Comp, Inc. Updated periodically. http://online.lexi.com/lco/action/doc/retrieve/docid/patch_f/7391/. Accessed November 5, 2014.
14. Olanzapine. DRUGDEX® System. Greenwood Village, CO: Thomson Healthcare. Updated periodically. http://www.micromedexsolutions.com/micromedex2/librarian/ND_T/evidencexpert/ND_PR/evidencexpert/CS/7BF70A/ND_AppProduct/evidencexpert/DUPLICATIONSHIELDSYNC/DC C3D4/ND_PG/evidencexpert/ND_B/evidencexpert/ND_P/evidencexpert/PFActionId/evidencexpert.DisplayDrugdexDocument?docId=923885&contentSetId=100&title=Olanzapine&servicesTitle=Olanzapine/. Accessed November 5, 2014.
15. Lilly Pharmaceuticals Inc. Instructions to reconstitute and administer Zyprexa Relprevv (website). 2014. <https://www.zyprexarelpvprogram.com/PDF/Lilly%20Recon%20and%20Admin%20Poster.pdf/>. Accessed November 11, 2014.
16. Risperdal Consta (risperidone) [package insert]. Titusville, NJ: Janssen Pharmaceuticals, Inc.; 2014.
17. Risperidone. Lexi-Drugs Online. Hudson, OH: Lexi-Comp, Inc. Updated periodically. http://online.lexi.com/lco/action/doc/retrieve/docid/patch_f/7632/. Accessed November 5, 2014.
18. Risperidone. DRUGDEX® System. Greenwood Village, CO: Thomson Healthcare. Updated periodically. http://www.micromedexsolutions.com/micromedex2/librarian/ND_T/evidencexpert/ND_PR/evidencexpert/CS/829D43/ND_AppProduct/evidencexpert/DUPLICATIONSHIELDSYNC/B8713C/ND_PG/evidencexpert/ND_B/evidencexpert/ND_P/evidencexpert/PFActionId/evidencexpert.DisplayDrugdexDocument?docId=922222&contentSetId=100&title=Risperidone&servicesTitle=Risperidone/. Accessed November 5, 2014.
19. Chue P, Eerdekens M, Augustyns I, Lachaux B, Molcan P, Eriksson L, et al. Comparative efficacy and safety of long-acting risperidone and risperidone oral tablets, Eur Neuropsychopharmacol. 2005;15(1):111-7. Invega Sustenna (paliperidone) [package insert]. Titusville, NJ: Janssen Pharmaceuticals, Inc.; 2014
20. Paliperidone. Lexi-Drugs Online. Hudson, OH: Lexi-Comp, Inc. Updated periodically. http://online.lexi.com/lco/action/doc/retrieve/docid/patch_f/778029/. Accessed November 5, 2014.

*Cost data representative of acquisition cost from AmerisourceBergen as of 11/11/2014

21. Paliperidone. DRUGDEX® System. Greenwood Village, CO: Thomson Healthcare. Updated periodically.
http://www.micromedexsolutions.com/micromedex2/librarian/ND_T/evidencexpert/ND_PR/evidencexpert/CS/3B29A0/ND_AppProduct/evidencexpert/DUPLICATIONSHIELDSYNC/C0E93D/ND_PG/evidencexpert/ND_B/evidencexpert/ND_P/evidencexpert/PFActionId/evidencexpert.DisplayDrugdexDocument?docId=928711&contentSetId=100&title=Paliperidone&servicesTitle=Paliperidone/. Accessed November 5, 2014.